

Safeband_X

GPIO Pin Assignment Map — Group WD025

All members must read and confirm their row before wiring. Do not connect any component without checking this table first.

Member Colour Key

M1 — Core Firmware	M2 — MPU6050	M3 — Audio / AVI	M4 — GPS / Blynk	M5 — Stream / Log	M6 — Hardware
--------------------	--------------	------------------	------------------	-------------------	---------------

GPIO Pin Assignments — Your Components

i The ESP32-CAM uses most of its GPIOs internally for the OV2640 camera and SD card. Only the pins listed below are free to use for your components. Do not use any pin not listed in this table.

GPIO 13	MPU6050 SDA (I2C data)	M2	I2C bus — pair with GPIO 15	Yellow	Confirm with M1 no other device uses SDA. Pull-up 4.7kΩ to 3.3V recommended on the line.
GPIO 15	MPU6050 SCL (I2C clock)	M2	I2C bus — pair with GPIO 13	White	Confirm with M1 no other device uses SCL. Pull-up 4.7kΩ to 3.3V recommended on the line.
GPIO 12	MPU6050 INT (interrupt)	M2	⚠ MUST be LOW at boot. Do NOT add a pull-up resistor here.	Orange	INT pin on MPU6050 is normally LOW and pulses HIGH on impact — this is correct. Never add a pull-up.
GPIO 33	Panic button	M1	INPUT_PULLUP in code. Button other leg to GND.	Red	Confirm GPIO 33 is not used by camera. It is free on AI Thinker ESP32-CAM.
GPIO 3 (RX0)	Safe cancel button	M1	⚠ Shares UART0 RX with Serial. Disable or use carefully when GPS active.	Blue	If GPS conflicts with this pin, move cancel button to GPIO 32 and update the team.
GPIO 14	Piezo siren (via 2N2222)	M6	Drive transistor base via 1kΩ resistor. Never direct connect siren to GPIO.	Purple	Test siren + transistor circuit separately before connecting to ESP32-CAM.
GPIO 2	Red LED (via 470Ω resistor)	M6	⚠ Also onboard flash LED. If SD card issues appear, move LED to GPIO 32.	White	Test LED independently first. If AVI recording corrupts, swap to GPIO 32 immediately.
GPIO 1 (TX0)	GPS Neo-6M RX	M4	⚠ UART0 TX — Serial.print() stops when	Green	Use HardwareSerial gpsSerial(1) with custom RX/TX pins. Discuss with M1 before wiring.

			GPS active. Switch to HardwareSerial(1).		
GPIO 26	INMP441 WS (I2S word select)	M3	I2S bus — all three INMP441 pins must be connected together.	Yellow	L/R pin on INMP441 must be tied to GND for left channel address 0.
GPIO 25	INMP441 SCK (I2S bit clock)	M3	I2S bus	White	Confirm I2S port used (I2S_NUM_0 or I2S_NUM_1) with M1 — only one I2S port available after camera.
GPIO 35	INMP441 SD (I2S data in)	M3	Input-only pin — correct for I2S data receive.	Orange	GPIO 35 is input only — correct for microphone data. Do not use for output.

Reserved Pins — Do NOT Use

⚠ The following pins are used internally by the ESP32-CAM for the OV2640 camera and SD card slot. Connecting anything to these pins WILL cause the camera, SD card, or the whole board to malfunction.

GPIO Pin(s)	Reserved For	Why — do not touch
GPIO 0	Boot / Camera	Must be LOW to flash, HIGH to run. The MB board handles this. Do not connect anything here.
GPIO 4	SD card / Flash LED	Onboard white flash LED is on GPIO 4. Also used by SD card. Avoid completely.
GPIO 5, 18, 19, 21, 22, 23	Camera SCCB / I2C	OV2640 camera uses all of these for its control bus. Connecting anything here kills the camera.
GPIO 27, 32, 34	SD card data lines	Used by the onboard SD card slot for data transfer. Do not use.
GPIO 36, 39	Input only / ADC	These are input-only, no internal pull-up, and share ADC2 with WiFi. Unreliable when WiFi is active. Avoid.

Power Connections — Same for All Members

ESP32-CAM Pin	Connect To	Notes
3.3V	MPU6050 VCC, INMP441 VDD, GPS VCC, LED anode (via resistor), button pull-up	All low-current sensors run on 3.3V. Do not connect sensors directly to 5V — they will be damaged.

GND	All component GND pins — MPU6050, INMP441, GPS, siren, LED, buttons, TP4056 GND	All GND pins must share a common ground rail on the breadboard.
5V (input)	TP4056 OUT+ (via boost converter if needed)	LiPo outputs 3.7V–4.2V. Try 3.3V pin first. Use MT3608 boost converter only if 5V pin is needed.

Critical Warnings for the Whole Team

- ⚠ GPIO 12 must be LOW at boot. Never add a pull-up resistor to GPIO 12. The MPU6050 INT pin is naturally LOW so this is fine — just don't change it.

⚠ The GPS Neo-6M and Serial.print() share the same UART pins (GPIO 1 and 3). When Member 4 activates GPS, Serial debug stops working. Member 4 must use HardwareSerial(1) with custom pins and discuss with Member 1 before wiring.

⚠ GPIO 2 is shared with the onboard SD card. If the red LED on GPIO 2 causes SD card corruption during AVI recording, Member 6 must immediately move the LED to GPIO 32 and update this document.

⚠ Never connect the piezo siren directly to a GPIO pin. It draws too much current and will damage the ESP32-CAM. Always use the 2N2222 NPN transistor circuit with a 1kΩ base resistor.
- ✓ Always test each component individually on a separate breadboard before connecting it to the main ESP32-CAM circuit. It is much easier to debug one component in isolation than to find a fault in a fully assembled circuit.

✓ When you change anything in this pin map, update this document immediately and send the new version to all 6 members on WhatsApp. Write your name and the date next to any change.

Member Sign-Off — Read Before Wiring

Each member must confirm they have read and understood their GPIO assignments before touching any wire. Fill in the date when confirmed.

Member	Name	Role	GPIO Pins Owned	Date Confirmed
M1	IT26100084	Core firmware, FreeRTOS, panic button	GPIO 33, GPIO 3 (cancel button)	
M2	IT26100156	MPU6050 impact detection	GPIO 13 (SDA), GPIO 15 (SCL), GPIO 12 (INT)	

M3	IT26100301	INMP441 audio, AVI recording	GPIO 26 (WS), GPIO 25 (SCK), GPIO 35 (SD)	
M4	IT26100942	GPS Neo-6M, Blynk notifications	GPIO 1 (TX), HardwareSerial(1) custom pins	
M5	IT26100297	MJPEG stream, EEPROM log	No additional GPIO — uses camera and WiFi from M1	
M6	IT26100757	Hardware assembly, siren, LED, power	GPIO 14 (siren), GPIO 2 (LED), TP4056 power rail	